FDE INTERVIEW PREP #05

FDE INTERVIEW PREP · ISSUE 05 OF 08

Data engineering questions for FDEs.

ETL/ELT, schema contracts, data quality, CDC, Parquet.

05

AR

Akhilesh Ranjan Singh

Senior DevSecOps Engineer · github.com/iarsingh

Swipe →

Questions and short answers

01 ETL or ELT?

ELT when the warehouse can transform. ETL when data must be cleaned first.

02 Schema contract?

Agreed fields and types, checked in CI and at ingest.

03 CDC or polling?

CDC for deletes and low latency. Polling for small daily syncs.

04 Why Parquet?

Columnar, typed, compressed, cheap to scan.

AR
Akhilesh Ranjan Singh
github.com/iarsingh

A pipeline you can defend

1. Ingest raw, unchanged
2. Validate against contract
3. Quarantine bad rows
4. Transform + test
5. Publish with freshness check

AR
Akhilesh Ranjan Singh
github.com/iarsingh

Quality checks that matter

CHECK	CATCHES
Row count vs yesterday	Missing or duplicate loads
Null rate per column	Upstream schema changes
Freshness	Stalled pipelines
Uniqueness of keys	Double processing

AR
Akhilesh Ranjan Singh
github.com/iarsingh

How to answer

DO

- ✓ Make loads idempotent
- ✓ Keep raw data replayable
- ✓ Alert on freshness
- ✓ Name who owns each source

AVOID

- ✗ Fail the whole batch on one row
- ✗ Silently drop bad rows
- ✗ Append on every retry
- ✗ Ignore deletes

AR
Akhilesh Ranjan Singh
github.com/iarsingh

Practise these

- [x] Design an incremental load
- [x] Write a schema contract
- [x] Explain CDC and WAL
- [x] Partition Parquet by date
- [x] Backfill without duplicates

AR
Akhilesh Ranjan Singh
github.com/iarsingh

FDE INTERVIEW PREP #05

KEY TAKEAWAY

Idempotent, validated and replayable is the bar.

→ Contracts catch upstream changes early

→ Quarantine, do not drop, bad rows

→ Every load safe to re-run

YOUR TURN

What breaks most often in your customer data pipelines?

FOLLOW AKHILESH RANJAN SINGH

Next in the series: #06 Cloud and DevOps questions for FDEs Save · Repost

AR
Akhilesh Ranjan Singh
github.com/iarsingh